

AIML SESSION 02 ASSIGNMENT SUBMISSION

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Domain/subject name: AIML (Artificial Intelligence & Machine Learning)

College and Branch name: Government Polytechnic College Washim (IF)

Task 1: if Statement

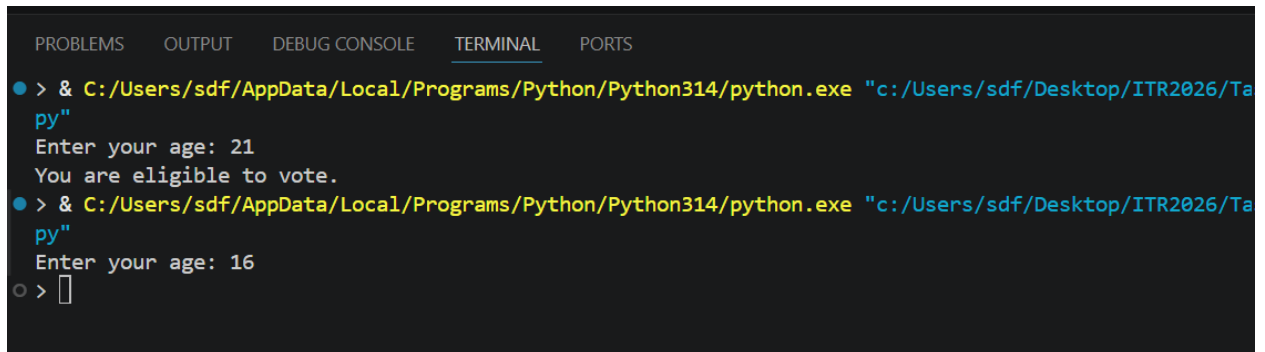
Program Code:

```
# Q1.py - Check voting eligibility

# Take age as input from user
age = int(input("Enter your age: "))

# Check if user is eligible to vote
if age >= 18:
    print("You are eligible to vote.")
```

OUTPUT Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Ta
py"
Enter your age: 21
You are eligible to vote.
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Ta
py"
Enter your age: 16
○ > 
```

Task 2: if-else Statement

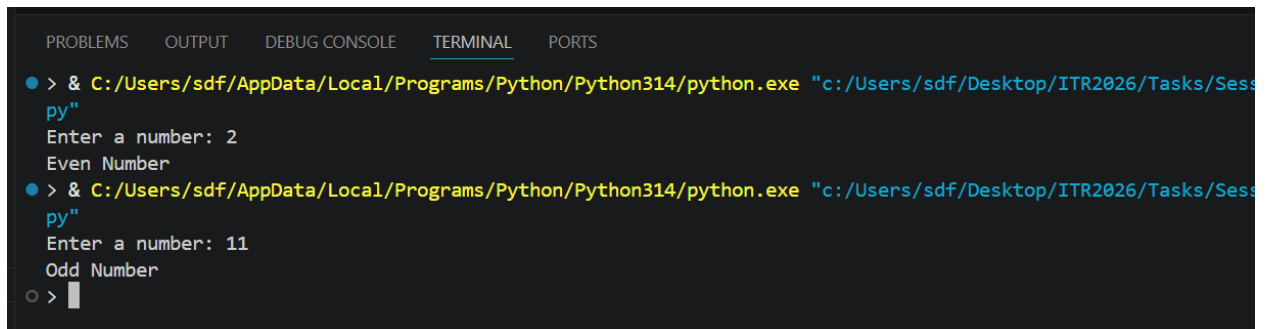
Program Code:

```
# Q2.py - Check even or odd number

# Take number as input
num = int(input("Enter a number: "))

# Check if number is divisible by 2
if num % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")
```

OUTPUT Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 2/Q2.py"
Enter a number: 2
Even Number
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 2/Q2.py"
Enter a number: 11
Odd Number
○ > █
```

Task 3: if-elif-else Statement

Program Code:

```
# Q3.py - Grade calculator based on marks

# Take student marks as input
marks = int(input("Enter your marks: "))

# Check marks against grading scale
if marks >= 90:
    print("Grade A - Excellent")
elif marks >= 75:
    print("Grade B - Good")
elif marks >= 60:
    print("Grade C - Average")
elif marks >= 40:
    print("Grade D - Pass")
else:
    print("Fail")
```

OUTPUT Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 3/Q3.py"
Enter your marks: 98
Grade A - Excellent
○ > |
```

Task 4: For Loop with range()

Program Code:

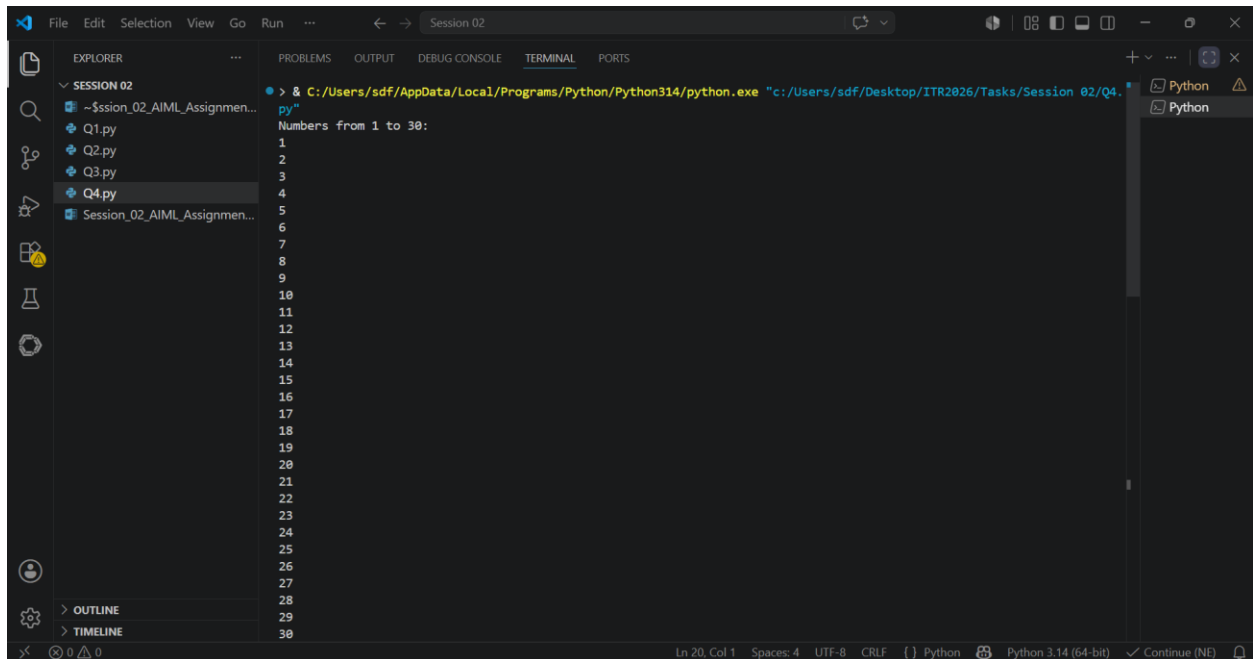
```
# Q4.py - For loop using range functions

# range(1, 31) generates values from 1 to 30
print("Numbers from 1 to 30:")
for i in range(1, 31):
    print(i)
print("-----")

# range(1, 51, 2) starts at 1, goes up to 50, increments by 2
print("Odd numbers from 1 to 50:")
for i in range(1, 51, 2):
    print(i)
print("-----")

# Calculating sum of numbers from 1 to 15
total = 0
for i in range(1, 16):
    total += i
print("Sum =", total)
```

OUTPUT Screenshot:



The screenshot shows a Python IDE with a dark theme. The Explorer panel on the left shows a project named 'SESSION 02' containing files 'Q1.py', 'Q2.py', 'Q3.py', and 'Q4.py'. The 'Q4.py' file is selected. The main editor window displays the code from the 'Program Code' block. The TERMINAL panel at the bottom shows the command prompt output: 'Numbers from 1 to 30:' followed by a list of numbers from 1 to 30, and '-----' followed by 'Odd numbers from 1 to 50:' followed by a list of odd numbers from 1 to 50, and '-----'. The status bar at the bottom indicates 'Ln 20, Col 1', 'Spaces: 4', 'UTF-8', 'CRLF', 'Python', and 'Python 3.14 (64-bit)'.

Task 5: While Loop

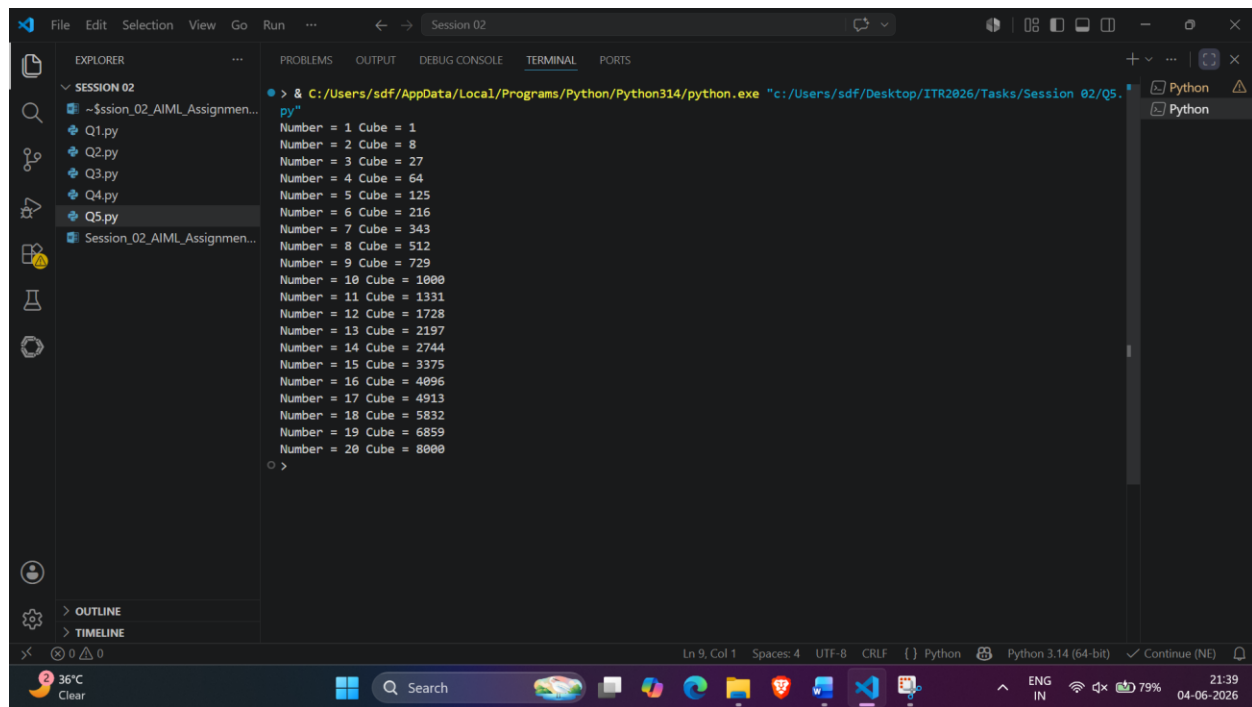
Program Code:

```
# Q5.py - Print numbers and cubes using while loop

num = 1

# Loop runs while num is 20 or less
while num <= 20:
    print("Number =", num, "Cube =", num ** 3)
    num += 1
```

OUTPUT Screenshot:



```
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 02/Q5.py"
Number = 1 Cube = 1
Number = 2 Cube = 8
Number = 3 Cube = 27
Number = 4 Cube = 64
Number = 5 Cube = 125
Number = 6 Cube = 216
Number = 7 Cube = 343
Number = 8 Cube = 512
Number = 9 Cube = 729
Number = 10 Cube = 1000
Number = 11 Cube = 1331
Number = 12 Cube = 1728
Number = 13 Cube = 2197
Number = 14 Cube = 2744
Number = 15 Cube = 3375
Number = 16 Cube = 4096
Number = 17 Cube = 4913
Number = 18 Cube = 5832
Number = 19 Cube = 6859
Number = 20 Cube = 8000
>
```

Task 6: While Loop - User Controlled

Program Code:

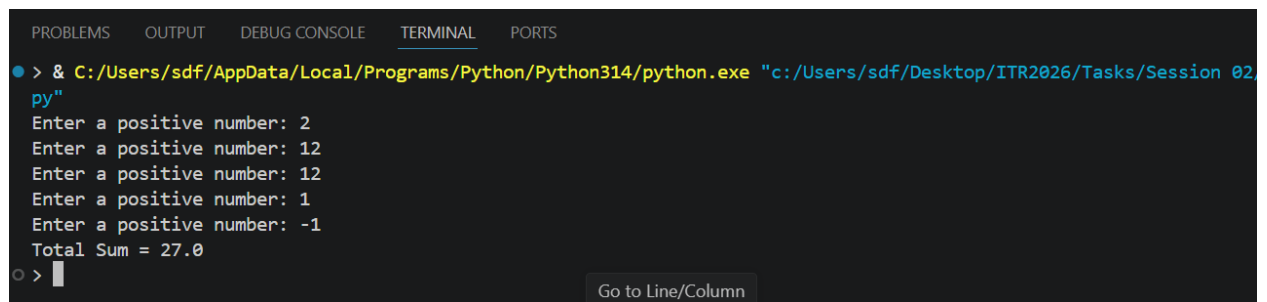
```
# Q6.py - Keep adding positive numbers until zero or negative is entered

total = 0.0

# Repeatedly take input until non-positive number is entered
while True:
    num = float(input("Enter a positive number: "))
    if num <= 0:
        break
    total += num

print("Total Sum =", total)
```

OUTPUT Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 02,
py"
Enter a positive number: 2
Enter a positive number: 12
Enter a positive number: 12
Enter a positive number: 1
Enter a positive number: -1
Total Sum = 27.0
○ > | Go to Line/Column
```

Task 7: Nested if Statement

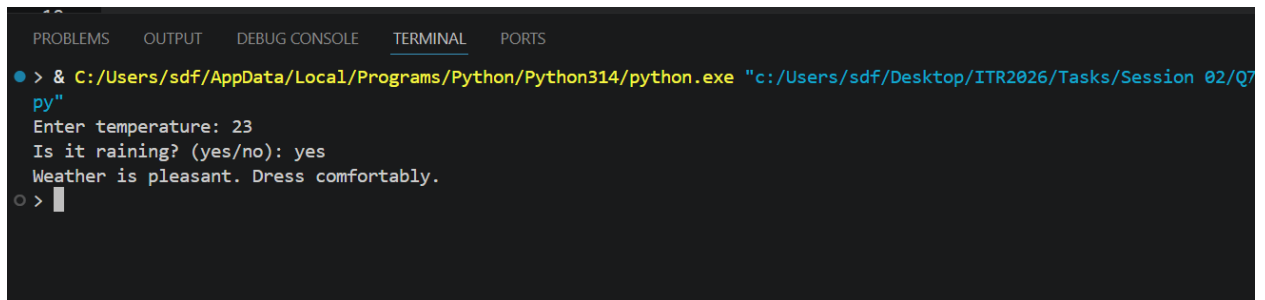
Program Code:

```
# Q7.py - Weather advice using nested if statements

temperature = int(input("Enter temperature: "))
is_raining = input("Is it raining? (yes/no): ")

# Nested logic for weather advice
if temperature > 30:
    if is_raining.lower() == "yes":
        print("Hot and rainy, carry umbrella.")
    else:
        print("Hot day, wear light clothes.")
elif temperature < 15:
    if is_raining.lower() == "yes":
        print("Cold and rainy, wear jacket and take umbrella.")
    else:
        print("Cold day, wear warm clothes.")
else:
    print("Weather is pleasant. Dress comfortably.")
```

OUTPUT Screenshot:



The screenshot shows a terminal window with the following content:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 02/Q7.py"
Enter temperature: 23
Is it raining? (yes/no): yes
Weather is pleasant. Dress comfortably.
○ > █
```

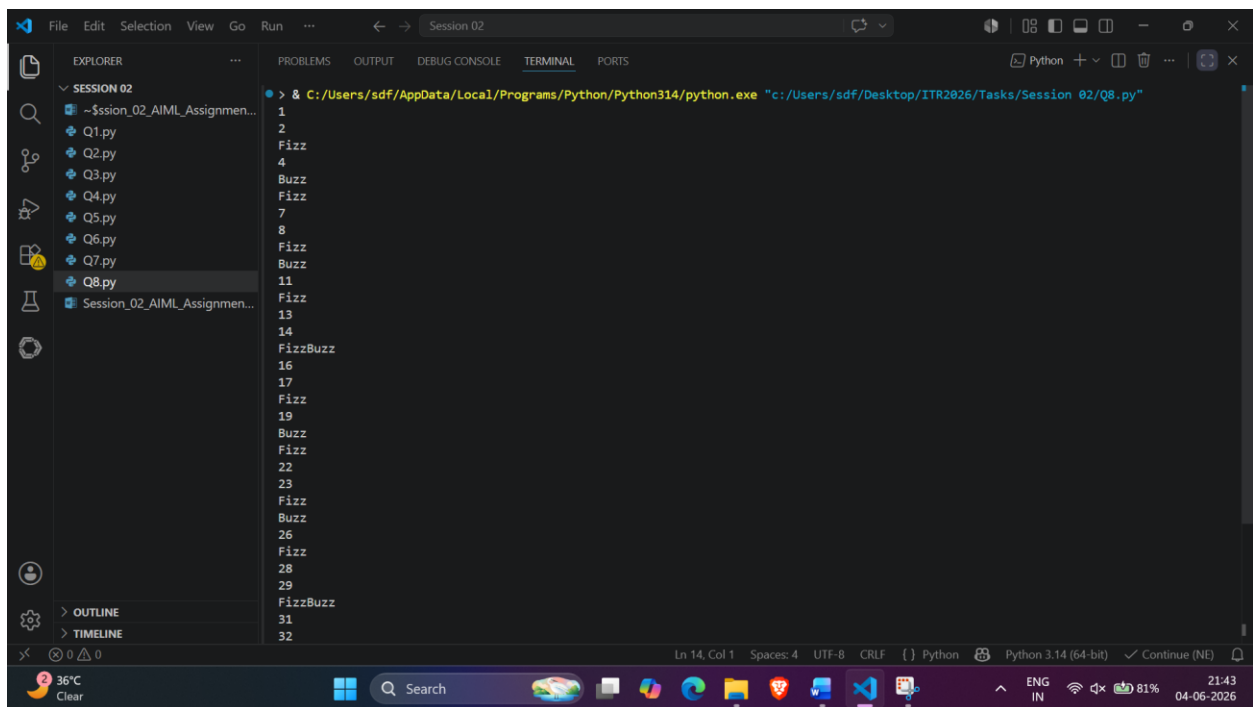

Task 8: For Loop + if-elif-else

Program Code:

```
# Q8.py - FizzBuzz from 1 to 40

# Loop through numbers 1 to 40
for num in range(1, 41):
    if num % 3 == 0 and num % 5 == 0:
        print("FizzBuzz")
    elif num % 3 == 0:
        print("Fizz")
    elif num % 5 == 0:
        print("Buzz")
    else:
        print(num)
```

OUTPUT Screenshot:



```
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 02/Q8.py"
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
Fizz
19
Buzz
Fizz
22
23
Fizz
Buzz
26
Fizz
28
29
FizzBuzz
31
32
```

Task 9: Menu Driven Program

Program Code:

```
# Q9.py - Menu driven calculator

while True:
    print("\nCalculator Menu")
    print("1. Add two numbers")
    print("2. Subtract two numbers")
    print("3. Multiply two numbers")
    print("4. Divide two numbers")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 5:
        print("Exiting Calculator...")
        break

    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))

    if choice == 1:
        print("Result =", num1 + num2)
    elif choice == 2:
        print("Result =", num1 - num2)
    elif choice == 3:
        print("Result =", num1 * num2)
    elif choice == 4:
        if num2 == 0:
            print("Error! Division by zero is not allowed.")
        else:
            print("Result =", num1 / num2)
    else:
        print("Invalid Choice")
```

OUTPUT Screenshot:

```
File Edit Selection View Go Run ...  
EXPLORER  
SESSION 02  
--$sion_02_AIML_Assignmen...  
Q1.py  
Q2.py  
Q3.py  
Q4.py  
Q5.py  
Q6.py  
Q7.py  
Q8.py  
Q9.py  
Session_02_AIML_Assignmen...  
OUTLINE  
TIMELINE  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
Python  
C:\Users\sdf\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 02/Q9.py"  
Calculator Menu  
1. Add two numbers  
2. Subtract two numbers  
3. Multiply two numbers  
4. Divide two numbers  
5. Exit  
Enter your choice: 1  
Enter first number: 12  
Enter second number: 6  
Result = 18.0  
Calculator Menu  
1. Add two numbers  
2. Subtract two numbers  
3. Multiply two numbers  
4. Divide two numbers  
5. Exit  
Enter your choice: 1  
Enter first number: 12  
Enter second number: 12  
Result = 24.0  
Calculator Menu  
1. Add two numbers  
2. Subtract two numbers  
3. Multiply two numbers  
4. Divide two numbers  
5. Exit  
Enter your choice:   
Ln 33, Col 1 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Continue (NE)  
36°C Clear 21:45 04-06-2026
```

Task 10: Debugging

Program Code:

```
# Q10.py - Debugged python code with fixes

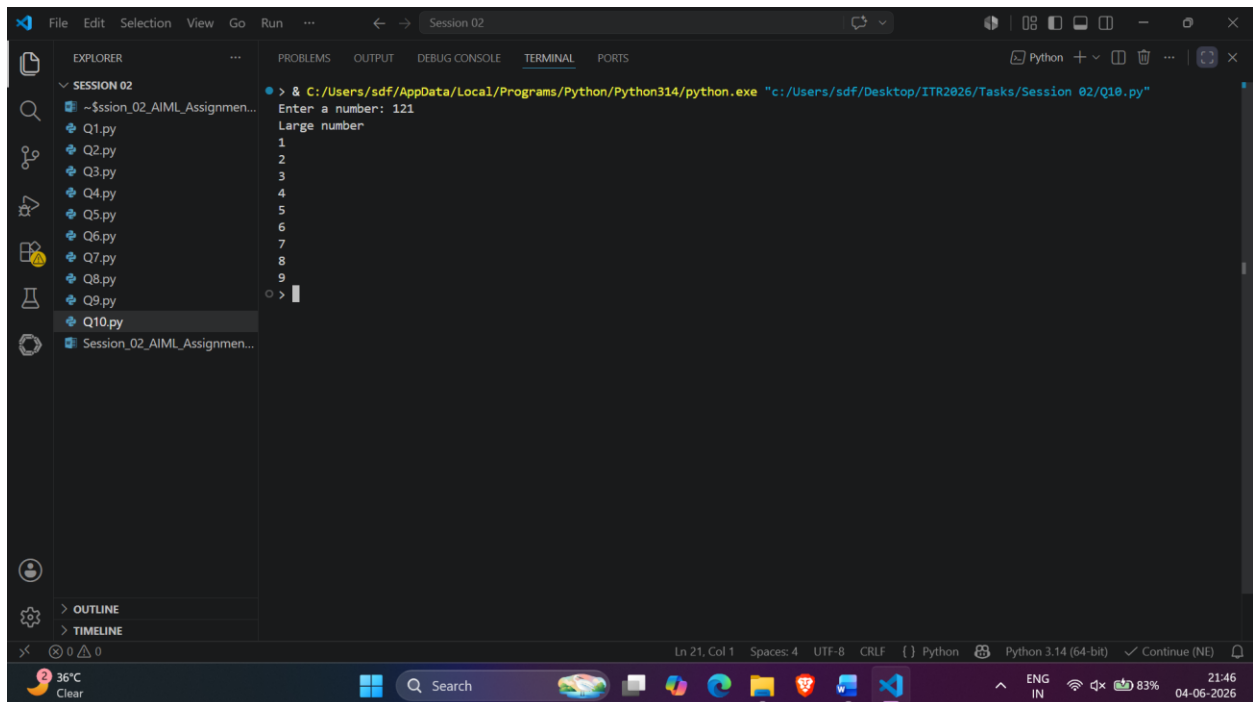
# Convert input to integer because input() returns a string
num = int(input("Enter a number: "))

# Added missing colons (:) to if-elif-else statements
if num > 100:
    print("Large number")
elif num > 50:
    print("Medium number")
else:
    print("Small number")

count = 1

# Added missing colon (:) to while statement
while count < 10:
    print(count)
    # Corrected count increment syntax
    count += 1
```

OUTPUT Screenshot:



The screenshot shows a Python IDE with the following components:

- EXPLORER:** A list of files including Q1.py through Q10.py and a folder named Session_02_AIML_Assignmen... Q10.py is selected.
- TERMINAL:** The command prompt shows the execution of the script: `C:\Users\sdf\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 02/Q10.py"`. The output is: `Enter a number: 121`, `Large number`, and a list of numbers from 1 to 9.
- STATUS BAR:** The bottom status bar indicates the current line and column (Ln 21, Col 1), the file encoding (UTF-8), the line ending (CRLF), the Python version (Python 3.14 (64-bit)), and the interpreter path (Continue (NE)).